

HAIDER ALI

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EDUCATION

Yeshiva University - NYC, USA - Master of Science, Artificial Intelligence **May 2024**
Mumbai University - Mumbai, India - Bachelor of Science, Computer Science, Specialization in AI **May 2021**

PUBLICATIONS

- **KatzBot**: Revolutionizing Academic Chatbot for Enhanced Communication **May 2024**
- **Cow Teat Keratosis Classification**: Data Augmentation Using Stable Diffusion for Imbalanced Biological Dataset **May 2023**

TECHNICAL SKILLS

- **Programming Tools**: Python, PyTorch, Semantic Kernel, FastAPI, Docker, JAX, Git, MySQL, PostgreSQL, HuggingFace, C++
- **Cloud & Data**: GCP, Vertex, Azure, Plotly, Vector DBs, Scikit-learn, Pandas, HIPAA, Matplotlib, CI/CD, LangChain, FAISS, EMR
- **ML/AI**: LLMs (GPT, BERT, OpenAI, Gemini), RAG, Prompt/Context Engineering, NLP, Computer Vision, Generative AI (Stable Diffusion, GAN), XGBoost, ONNX, FlashAttention, LLM Fine-tuning (QLoRA, Quantization, SFT), Vector search (sparse, hybrid), Tool calling, model evaluations, MLFlow, Transformers, MLOPS, Computer Vision, vLLM, FSDP, Langsmith, Pydantic

EXPERIENCE

Credo Health – *Machine Learning Engineer; Austin, TX* **June 2024 – Feb 2026**

- Built a hybrid **re-ranking RAG** pipeline serving concurrent multi-user clinical queries with **real-time document ingestion**
- Reduced LLM prompt tokens by **30% via hash-based deduplication**, lowering per-query inference **cost at scale**
- Deployed an agentic chatbot framework achieving **96% accuracy**, validated across thousands of QA test cases
- Designed a **multi-judge LLM eval** framework running **automated quality** assessments across all **production agent outputs**

S&P Global – *Machine Learning Joint Research Project; NYC, NY* **May 2023 – August 2023**

- Scraped, engineered, and extracted relevant features of Global Precipitation Data from NASA's GDDP (**HDF**) files using **R**
- Employed predictive modeling using features from KNN and CNN after morphological image analysis
- Evaluated multiple super-resolution models; **ESRGAN** maintained 80% image quality while enhancing resolution

Kan Innovations – *Machine Learning Engineer; Mumbai, India* **January 2022 – Feb 2023**

- Developed foot imagery dataset using scrapers and pressure mat data, processed with Pix2Pix GANs
- Leveraged **DenseNet121**, **DeepLabV3+**, and custom segmentation model to estimate the pressure points in a foot
- Constructed 3D voxel foot meshes using **Open3D** library and Blender to help design insoles, reduced **40%** foot deformity
- Implemented **BlazePose** model for real-time pose estimation, enhancing gaming interaction reducing **20%** foot deformity

PROJECTS

Credit card assistant | github.com/AliHaider20/Semantic-kernel-card-assist-agent | **Prompt engineering** **February 2025**

- Developed agentic AI chatbot using Semantic Kernel with tool-calling for card operations and document Q&A
- Built RAG pipeline ingesting PDFs, chunking via LangChain, and indexing with FAISS for grounded responses
- Built reusable tool/skill abstractions with clear input/output interfaces for card operations and document retrieval

University Chatbot using LLM | github.com/AliHaider20/KatzBot-LLM | **pre-training LLMs** **December 2023**

- Scraped, and synthetically generated 60K question-answer pairs and human using the **NLPAug** tool, **BERT**, and **GPT** models
- Fine-tuned **Phi 1.5B**, **Flan T5**, **GPT 2**, and **Mistral** models using FSDP, QLoRA, SFT, and GRPO achieving 70% RougeL score
- Instruction trained, and deployed a custom SLM model on local server using Streamlit and FastAPI

Lung Image Segmentation | github.com/AliHaider20/Lung-segmentation | **Paper** **October 2023**

- Applied **multi-Otsu thresholding** for image preprocessing and segmentation augmentation on clinical lung CT scan datasets
- Developed a custom 6-layered CNN achieving **80% IOU score**, within **5% of DeepLabV3** and **2% of UNet** benchmarks
- Tracked experiments with **MLflow** and leveraged **PyTorch Lightning** for efficient model training, evaluation, and deployment

ACHIEVEMENTS

- 1st place, GreyAtom Data Science Competition; top performer in 2 consecutive **Python Hackathons** (700+ participants)